

COURSE PLAN

BLOCKCHAIN TECHNOLOGY

Open Elective

Prepared by

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UNIT-WISE COURSE PLAN

BLOCKCHAIN TECHNOLOGY

Course Type: Open Elective | **Total Hours:** 60 (15 weeks × 4 hours)

Software / Tools: Block explorer, Remix IDE, MetaMask, Ganache / Truffle, Node.js with Web3.js / Ethers.js

Text Books: (1) Imran Bashir, *Mastering Blockchain*, 2/e, Packt, 2017; (2) Narayanan, Bonnie, Felten, Miller, Goldfeder, *Bitcoin and Cryptocurrency Technologies*, Princeton University, 2016; (3) Subramanian, George, Abhilash, Karthikeyan, *Blockchain Technology*, University Press, 2021

UNIT I: Block Chain and its History — 10 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Course introduction; History of blockchain – from digital cash to Bitcoin; need for blockchain	1	Lecture / PPT	Timeline of key milestones in blockchain history
2	Types of blockchain – public, private, hybrid; comparison with traditional databases	1	Lecture / Board	Compare blockchain types with real-world examples
3	Blockchain Components – blocks, transactions, nodes, peer-to-peer network, ledger	1	Lecture / Board	Label the components of a blockchain network
4	Cryptographic primitives – hash functions, public-key cryptography, digital signatures	2	Lecture / Demonstration	Compute SHA-256 hashes; sign and verify a message
5	Permissioned Blockchain, Permissionless Blockchain and Consortium Blockchain	2	Lecture / Case Study	Classify Hyperledger, Bitcoin, Ethereum and R3 Corda
6	Basics of Consensus Algorithms – need for consensus, Byzantine Generals problem	1	Lecture / Think-Pair-Share	Discuss fault tolerance in distributed agreement
7	Architecture and Properties of Blockchain – immutability, transparency, decentralisation, security	2	Lecture / Worked Examples	Draw layered blockchain architecture; quiz on Unit I

UNIT II: Decentralization and Consensus Algorithms — 13 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Decentralization using blockchain; Methods of decentralization – disintermediation, contest-driven	2	Lecture / Board	Identify intermediaries removed by blockchain in given scenarios
2	Routes to decentralization	1	Lecture / Discussion	Map the steps to decentralize a given system
3	Decentralized organizations – DOs, DAOs, DACs, DApps	1	Lecture / Case Study	Study The DAO case and its lessons
4	Distributed systems – CAP theorem, fault tolerance	1	Lecture / Worked Examples	Apply the CAP theorem to example systems
5	Distributed ledger – centralized vs. decentralized vs. distributed ledgers	1	Lecture / Board	Compare ledger models
6	Merkle tree – construction, Merkle root, Merkle proofs	1	Demonstration / Problem Solving	Build a Merkle tree for a set of transactions
7	Structure of a block – block header, nonce, timestamp, previous hash, genesis block	1	Demonstration	Inspect real blocks using a block explorer
8	Consensus Algorithms: Proof of Work – mining, difficulty adjustment, energy cost	2	Lecture / Demonstration	Simulate mining with varying difficulty
9	Proof of Stake and Proof of Burn	1	Lecture / Worked Examples	Compare PoW, PoS and PoB on cost and security
10	Proof of Elapsed Time and Proof of Activity	1	Lecture / Worked Examples	Explain PoET with trusted execution environments
11	Proof of Concept; comparison of consensus algorithms	1	Problem Solving / Quiz	Tabulate consensus algorithms; quiz on Unit II

UNIT III: Bitcoin and Alternative Coins — 13 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Bitcoin – overview, keys and addresses, wallets	1	Lecture / Demonstration	Create a test wallet and address
2	Bitcoin Transactions – inputs, outputs, UTXO model, transaction scripts	2	Lecture / Demonstration	Trace a transaction on a block explorer
3	Bitcoin payments; Bitcoin properties	1	Lecture / Board	List properties with supporting examples
4	Transaction life cycle	1	Lecture / Worked Examples	Draw the transaction life cycle
5	Creation of coin – mining reward, halving; Sending payments	1	Lecture / Worked Examples	Calculate block rewards across halving periods
6	Double spending using blockchain	1	Lecture / Think-Pair-Share	Analyse a double-spend attack scenario
7	Bitcoin anonymity – pseudonymity, mixing, de-anonymisation	1	Lecture / Discussion	Discuss privacy risks of public ledgers
8	Ether – Ethereum properties; comparison of Ether with Bitcoin	1	Lecture / Case Study	Compare Bitcoin and Ether features
9	Alternative Coins; Bitcoin limitations – scalability, energy, privacy	1	Lecture / Discussion	Discuss Bitcoin limitations and solutions
10	Namecoin and Litecoin	1	Lecture / Case Study	Compare with Bitcoin by purpose and consensus
11	Primecoin and Zcash	1	Lecture / Case Study	Study proof-of-work on primes and zero-knowledge proofs
12	Future currencies – CBDCs, stablecoins; creating own crypto token (ERC-20 overview)	1	Demonstration / Quiz	Outline the design of a custom token; quiz on Unit III

UNIT IV: Ethereum and Smart Contracts — 13 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Ethereum Architecture – EVM, accounts, transactions	2	Lecture / PPT	Explore accounts and transactions on a test network
2	Ethereum gas, blocks, networks and clients	1	Lecture / Demonstration	Estimate gas costs for sample transactions
3	Solidity programming basics – data types, variables, functions, control structures	2	Demonstration / Hands-on	Write simple Solidity programs in Remix IDE
4	Solidity – mappings, structs, modifiers, events, inheritance	1	Demonstration / Hands-on	Extend programs with modifiers and events
5	Smart Contract – structure, life cycle, common security issues	2	Lecture / Worked Examples	Write a simple storage and voting contract
6	Deploying Smart Contracts – Remix, MetaMask and test networks	2	Demonstration / Hands-on	Deploy a contract to a test network
7	Deploying Smart Contracts with Ganache and Truffle	1	Demonstration / Hands-on	Deploy and test a contract on a local blockchain
8	Integration with UI – Web3.js / Ethers.js, building a simple DApp front end	2	Demonstration / Mini Project	Connect a web page to a deployed contract; assignment

UNIT V: Blockchain Applications — 8 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Blockchain outside of currencies; Internet of Things	1	Lecture / Case Study	Study blockchain for IoT device management
2	Government	1	Lecture / Case Study	Discuss e-governance and land records
3	Health	1	Lecture / Case Study	Study sharing of electronic health records
4	Finance	1	Lecture / Case Study	Study cross-border payments, trade finance and DeFi
5	Media	1	Lecture / Discussion	Discuss content ownership and royalties
6	Secure Voting and Digital Identity	1	Lecture / Think-Pair-Share	Design a blockchain-based voting system
7	Real Estate	1	Lecture / Case Study	Study property registration and tokenisation
8	Education	1	Lecture / Case Study	Study certificate verification; quiz on Unit V

UNIT VI: Revision and Assessment — 3 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Revision of Units I–III – blockchain basics, consensus, Bitcoin and altcoins	1	Tutorial / Discussion	Solve previous question papers
2	Revision of Units IV–V – Ethereum, smart contracts and applications	1	Tutorial / Discussion	Solve previous question papers
3	Assignment review, slip test and doubt clearance	1	Slip Test / Discussion	Written test and feedback on assignments

Overall Hour Distribution

Unit	Unit Title	Hours	Percentage
I	Block Chain and its History	10	16.67%
II	Decentralization and Consensus Algorithms	13	21.67%
III	Bitcoin and Alternative Coins	13	21.67%
IV	Ethereum and Smart Contracts	13	21.67%
V	Blockchain Applications	8	13.33%
VI	Revision and Assessment	3	5.00%
	GRAND TOTAL	60	100%

General Modes of Teaching

- Each session is a 1-hour lecture: concepts are introduced on the board or through PPT, followed by examples and discussion.
- Live demonstrations with block explorers, hashing tools, Remix IDE and MetaMask make abstract concepts concrete.
- Hands-on sessions on Solidity programming and smart-contract deployment on Ethereum test networks.
- Case studies on The DAO, altcoins and blockchain in government, health, finance and education connect theory with practice.
- Active-learning techniques such as Think-Pair-Share and short quizzes check understanding at the end of each unit.
- Continuous assessment through assignments, quizzes, a mini project (simple DApp), slip test and mid-semester examinations.
- Additional reading and exercises are given to fast learners, and remedial sessions are held for slow learners.
- The final three hours are for revision, solving previous question papers and the slip test.